Compute the derivative of each of the following functions using the definition of the derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

State the domain of the function and the domain of its derivative.

1. $f(x) = \sqrt{x}$

2. $f(x) = \dfrac{1}{x}$

3. $f(x) = \sqrt{9-x}$

4. $f(x) = x^3$ (*Hint*: Recall that $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.)

5. $f(x) = x^4$ (*Hint*: Recall that $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.)

6. $f(x) = x^n$, where $n \in \mathbb{N}$.
(*Hint*: Recall that $(a+b)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \cdots + nxh^{n-1} + h^n$, where $\binom{n}{k} := \frac{n!}{k!(n-k)!}$ with $0! := 1$.)